Poster: SurakshaNet: Privacy-First, Edge-Deployed Multilingual Abusive Language Detection

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ABSTRACT

Cloud moderation APIs raise privacy concerns and add network latency. *SurakshaNet* is a Chrome extension that checks incoming WhatsApp Web messages on the device with an INT8-quantized multilingual MiniLM running in ONNX Runtime Web; no message text leaves the browser. INT8 quantization makes the model 75% smaller (470.6 MB to 118.4 MB) with no measurable accuracy loss, and a rolling-window gate lifts detection of abuse spread over several messages from 28.9% to 76.1% on a Hinglish test set. To study abuse in context, we generate a 105,000-thread dataset in English, Hindi and Hinglish with five classes (benign, friendly sarcasm, hostile abuse, sexual harassment, predatory grooming) and train thread-aware classifiers: on the held-out synthetic test set, the 118 MB INT8 model reaches 97.9% accuracy, about 15 points above a model that sees only the last message. We describe the generation pipeline, training and metrics in full.

CCS CONCEPTS

• **Security and privacy → Software and application security**.

KEYWORDS

Abusive language detection, Privacy-preserving AI, Edge computing, Synthetic data, Conversation threads

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1 INTRODUCTION

Abusive messages are common on social platforms [8], and women and children are frequent targets. In code-mixed Indian settings (Hindi, Hinglish) [5], sending private chats to a cloud service is often not acceptable. We therefore build detection that runs inside the browser. This paper makes three contributions.

(1) **On-device detector (Sections 2–3).** A Chrome extension that scores incoming WhatsApp Web messages locally with a 118 MB INT8 MiniLM, plus a rolling window that handles abuse spread over several messages.
(2) **Five-class thread dataset (Section 4).** A documented pipeline that generates 105,000 threads in three scripts, in which the label of the last message depends on the earlier messages.
(3) **Thread-aware models (Sections 5–6).** Three model variants, their training recipe, and their results, including an INT8 model of the same 118 MB size.

2 SYSTEM IMPLEMENTATION

Figure 1 shows what runs in the browser. A content script watches the WhatsApp Web page and passes each new incoming message and its chat name to a Manifest V3 service worker. The worker keeps a rolling window of at most 8 turns and 96 content tokens per chat and scores the message with the INT8 MiniLM through ONNX Runtime Web (WebAssembly). If the chat has earlier turns and the new message is short (fewer than 8 content tokens), the worker also scores the joined window and keeps the larger score. A message is flagged when the score exceeds 0.5 and marked *high* at 0.9 or above. Flagged messages get a warning banner and are saved to a local incident log (latest 200 incidents) that can be exported as JSON or PDF. Models load from local files only, and the inference path makes no network request.

3 ON-DEVICE DETECTOR

Model and data. The shipped detector is a multilingual MiniLM-L12 (`paraphrase-multilingual- MiniLM-L12-v2`) [10, 12] fine-tuned for 3 epochs on 58,929 messages: the Hindi part of MACD [5] (20,183), Davidson English tweets [3] (19,826) and a Hinglish set (18,920) obtained by converting MACD Hindi comments into Roman-script, code-mixed text with a Qwen-based converter. The held-out test set has 15,575 messages.

INT8 quantization. Dynamic INT8 quantization with ONNX Runtime [6] reduces the model from 470.6 MB to 118.4 MB (Figure 2, left). Accuracy is unchanged within 0.15 points on every test set (centre): 85.1% on Hindi, 84.4% on Hinglish, 96.7% on English and 86.5% on the mixed set. On Hindi this is within about one point of the 86.34% reported for an XLM-R model pretrained on MACD [5], with a much smaller model. A native single-thread run of the INT8 file takes a median of 5.0 ms per message including tokenization.

Context window. Abuse is often spread over several messages. On a Hinglish slice of 1,500 split-narrative examples, scoring only the current message finds 28.9% of them. Scoring the whole chat

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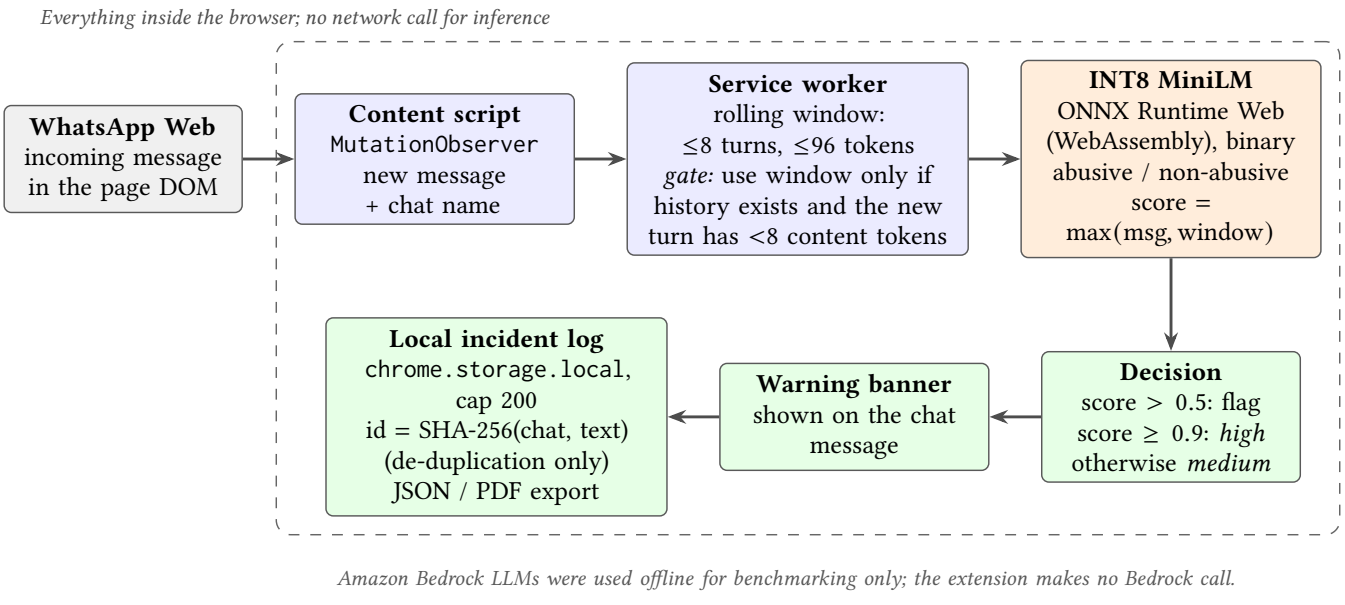


Figure 1: On-device pipeline of the extension. All blocks inside the dashed box run in the user's browser. Cloud LLMs were used offline for benchmarking only.

finds 79.3%, and the shipped gate finds 76.1% while keeping accuracy on abuse after small talk at 82.9% (right panel). On small subsets (20–80 examples) scored with two Amazon Bedrock LLMs, the local model matched or exceeded Nova Micro and Qwen3 32B on split-narrative recall (77.5% vs. 40.0% and 65.0%) without sending any text to a cloud service.

4 FIVE-CLASS THREAD DATASET

4.1 Why a new dataset

Binary datasets classify one message at a time, yet the same line ("try using your head") can be friendly teasing or a real attack depending on what came before (Figure 3), and grooming or harassment is defined by who is talking to whom. We therefore built a dataset in which the *label belongs to the last message given the earlier ones*. It has five classes (Table 1) chosen from prior taxonomies [1, 4, 7, 11, 13] and aimed at women and children.

4.2 Generation pipeline

Figure 4 gives the full flow. A grid of 3 scripts (English, Hindi in Devanagari, Hinglish in Roman letters), 4 lengths (3, 5, 7, 10 messages) and 5 classes gives 60 cells of 1,750 threads each, 105,000 in total. For every thread a seeded sampler picks a class-specific subtype (31 in total), a topic (40), two names from a pool of 30, the audience and, where relevant, the target's gender and the child's age. A prompt built from this scenario asks Qwen3-32B [9] on Amazon Bedrock to return exactly k alternating messages as JSON, where the last message follows the class rule. Rule-based validators then check the turn count, refusals, turn length, duplicates, script, and profanity or explicit words outside the hostile class, and sexualised words in grooming threads. In the last generation round, 98.6% of attempts passed. Accepted threads are de-duplicated and split 80/10/10 (83,672 train, 10,765 validation, 10,563 test).

Same-last-line pairs. About half of the friendly_sarcasm and hostile_abuse threads end in an identical line from a bank of 36 lines per script, so only the earlier messages carry the label (Figure 3). Pair lines are split by line, so no pair line appears in two splits; the test set holds 2,337 such threads.

Record format. Each record has 17 columns (Table 2). The dataset is balanced: 21,000 threads per class, 35,000 per script and 26,250 per length.

5 MODELS AND TRAINING

Figure 5 summarises the experiments. All models use the multilingual MiniLM-L12 encoder [10, 12] with a 5-way output.

Thread model (A). The input is the last message followed by the earlier messages, most recent first, separated by `[TURN]` (256 tokens maximum). It is fine-tuned for 2 epochs (learning rate 4×10^{-5}, batch 32, fp16) with the word-embedding table frozen, on one NVIDIA T4 GPU (AWS g4dn.xlarge, 19.8 minutes, $0.18), and then exported to INT8. **Last-message-only model (B)** uses the same recipe but sees only the final message. **Frozen encoder + GRU (C)** encodes each line once (prefix `in:` or `out:`, 384 numbers), runs a one-layer GRU (hidden size 128) over the vectors, and has only 198,021 trainable weights.

Metrics. *Accuracy* is the share of correctly classified test threads; *macro-F1* is the mean of the five per-class F1 scores. We report both overall and by thread length, script and class.

6 RESULTS

All numbers are on the 10,563 held-out synthetic test threads.

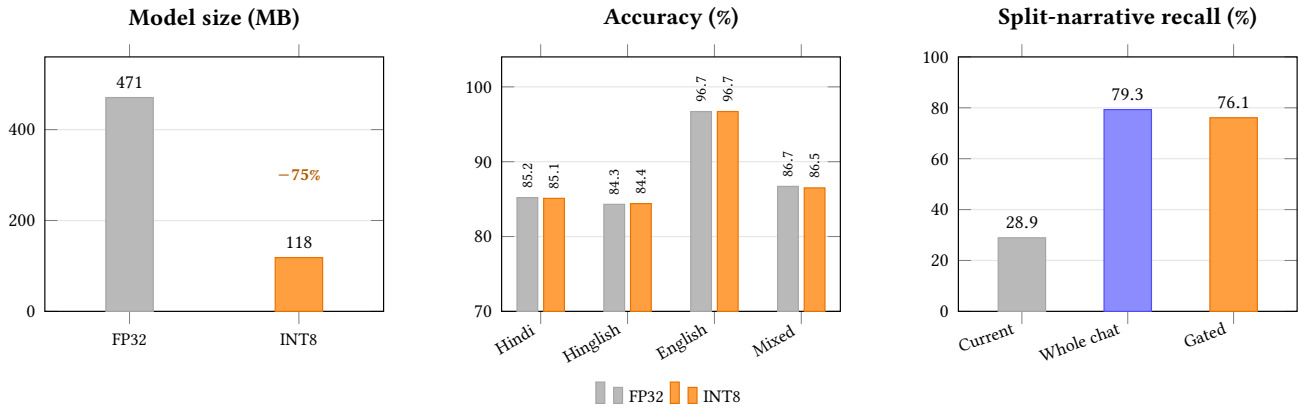


Figure 2: On-device detector. Left: INT8 quantization cuts the model size by 75%. Centre: test accuracy of the binary model before and after quantization. Right: share of split-narrative abuse detected with the current message only, the whole chat, and the gated window shipped in the extension (Hinglish test set).

Table 1: The five classes. The last-message rule is what the generation prompt asks for.

Id and class	Audience	Last-message rule in the prompt	Basis
0 benign	peers, adults, minors	Normal message. Includes hard negatives: parent–child, teacher–student, harmless surprise secrets, consensual flirting.	HASOC non-offensive [7]
1 friendly_sarcasm	peers	Playful, clearly affectionate teasing; the friends are laughing.	Van Hee et al. [11]
2 hostile_abuse	peers, adults	Hostile personal insult meant to hurt. No slurs about caste, religion, gender, ethnicity or disability; no sexual content.	Waseem et al. [13], MACD
3 sexual_harassment	adult woman (75%) or man (25%)	Unwanted suggestive remark or pressure; not graphic; the target deflects earlier.	Van Hee et al., Guest et al. [4]
4 predatory_grooming	minor (age 10–15)	Secrecy from parents, moving to a private app, meeting alone, age probing. No sexual content.	Banko et al. [1]

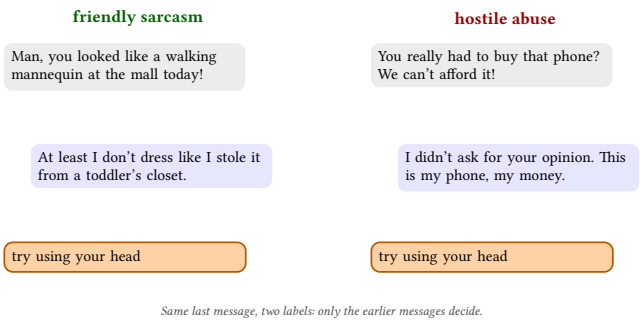


Figure 3: Two generated English threads with the same last message. The earlier messages decide whether it is friendly sarcasm or hostile abuse.

Table 2: The 17 columns of a record (`jsonl`).

Column	Meaning
`thread_id`, `split`	unique id; train / val / test
`length`, `n_context`	messages in the thread (3/5/7/10); earlier messages
`script`	english / hindi_devanagari / hinglish
`label`, `label_id`	one of the five classes; 0–4
`subtype`, `topic`	scenario (31 subtypes, 40 topics)
`audience`	peers / adult / minor, with gender where relevant
`target_gender`, `child_age`	woman/man or girl/boy; age 10–15 (grooming)
`pair_id`	set for same-last-line threads
`speakers`	the two speaker names, alternating
`turns`, `last_turn`	the messages in order; the final message
`source`	generator tag (Qwen3-32B via Bedrock, synthetic)

Thread awareness. Seeing the earlier messages lifts accuracy from 83.2% to 98.0% (Table 3), and the thread model stays above 96% at every length (Figure 6, left), while the last-message-only model gets worse as threads grow. On the same-last-line threads, which a last-message-only model cannot separate by construction, the INT8 thread model is correct 97.5% of the time.

Compact model. INT8 quantization shrinks the thread model by 75% for a loss of 0.08 points. The frozen-encoder variant needs only 198,021 trained weights and reaches 94.9%, which makes it attractive when training budget is small. Per script the INT8 thread

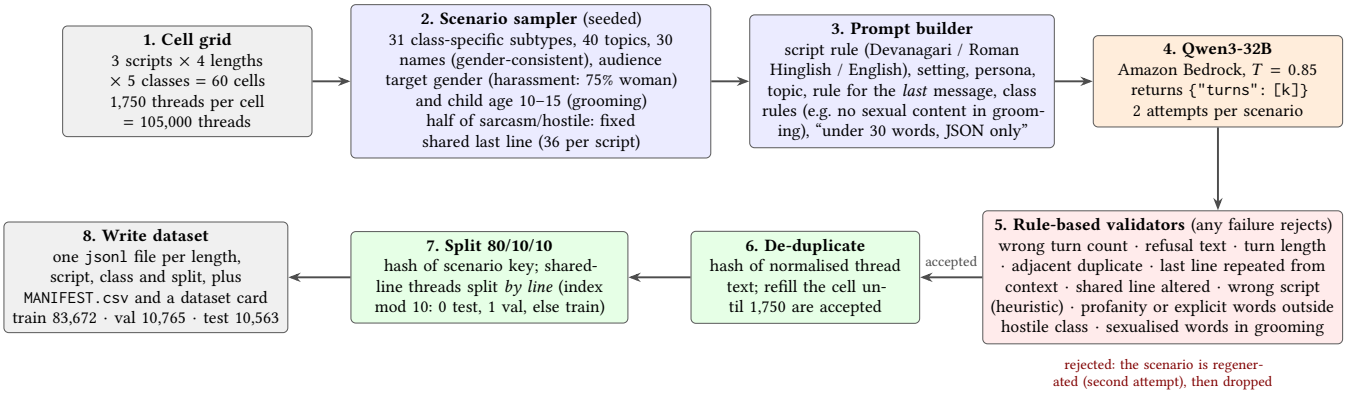


Figure 4: How the dataset was generated. Steps 1–4 create a scenario and a thread; step 5 rejects threads that break a rule; steps 6–8 de-duplicate, split and write the dataset.

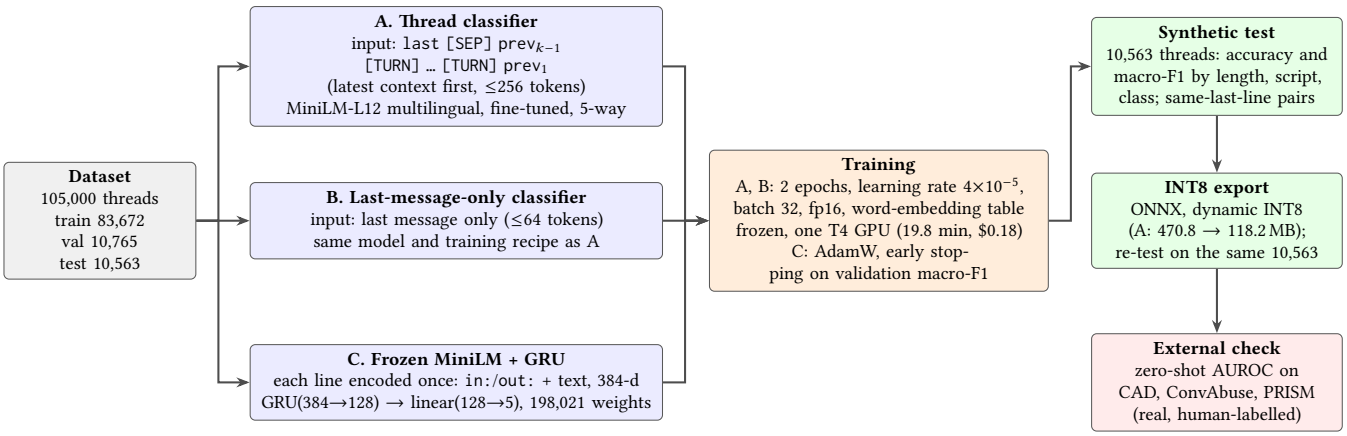


Figure 5: Models, training and evaluation. A and B are fine-tuned end to end; C keeps the encoder frozen and trains only a GRU [2] over per-line vectors.

Table 3: Five-class results on the synthetic test set.

Model	Size	Acc. (%)	Macro-F1
A. Thread model (FP32)	470.8 MB	97.96	0.980
A. Thread model (INT8)	**118.2 MB**	**97.88**	**0.979**
C. Frozen encoder + GRU (INT8)	118 MB + 0.8 MB	94.91	0.950
B. Last message only	470.8 MB	83.18	0.828

model scores 99.3% on English, 97.5% on Hindi and 96.9% on Hinglish, and per-class recall lies between 96.3% and 99.8% (Figure 6, right). With a native single-thread run, a 5-message thread takes a median of 28.3 ms.

7 LIMITATIONS AND NEXT STEPS

The five-class data are generated by an LLM and checked by automatic rules; no human annotators reviewed them, so the scores above describe the synthetic distribution. A first zero-shot check of the synthetic-trained models on real conversation data (CAD, ConvAbuse, PRISM) gave AUROC between 0.47 and 0.74, so we see the dataset today as a controlled benchmark and as training material to be mixed with real data. Our Hinglish text is machine-converted and less code-mixed than human Hinglish, the cloud-LLM comparison uses small subsets, and latency was measured natively rather than in the browser. The five-class models are not yet part of the extension, which uses the binary model. Next steps are: balancing the generator so that short last lines carry no label signal, adding real threaded data with held-out evaluation, obtaining human Hinglish threads (HASOC ICHCL), and measuring the INT8 models in the browser. The dataset contains synthetic abusive, harassing and grooming text and no real people; grooming threads use only non-sexual cues, and the data should be used only to build protective classifiers.

8 CONCLUSION

SurakshaNet shows that a 118 MB INT8 multilingual model can check WhatsApp Web messages entirely on the device, that a simple window gate recovers abuse spread across messages, and that

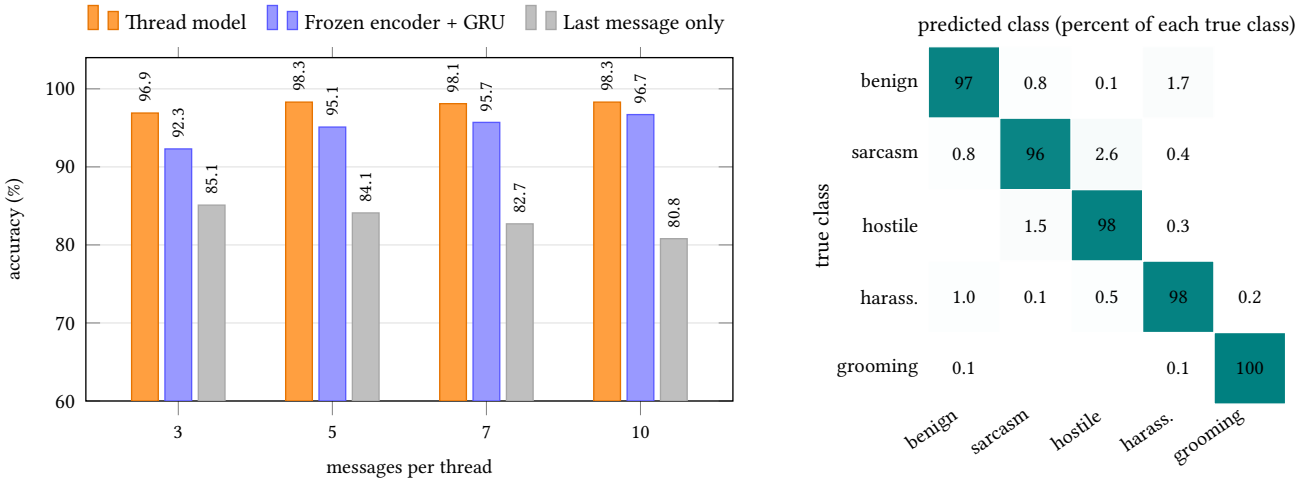


Figure 6: Five-class results on the synthetic test set. Left: accuracy by thread length. Right: confusion matrix of the INT8 thread model (row-normalised, percent).

a five-class thread benchmark lets us build models that use conversation context, reaching 97.9% on held-out synthetic threads with an INT8 model of the same size. Code, generation scripts and evaluation reports are in the project repository[1].

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